Matthew Emmanuel T. Labrador

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EDUCATION

University of the Philippines Manila *Manila, PH*
BS Computer Science — University Scholar | Expected May 2029
Current Cumulative GWA: 1.0375

PAREF Southridge School *Manila, PH*
Senior High School — Class Valedictorian, Gold Medalist | May 2025

EXPERIENCE AND PROJECTS

Diabetic Complication Swarm (GlycoSwarm AI) — AMD Developer Hackathon ACT II, Track 3 *2026*
Python, LangGraph, FastAPI, Next.js, Ollama, AMD MI300X GPU

- Led an international, cross-timezone team to design and build a multi-agent clinical decision-support prototype that screens already-diagnosed diabetic patients for renal, neuropathy, retinal, and cardiovascular complication risk.
- Team's system used a LangGraph StateGraph to run four parallel specialist agents, each generating and executing its own Python scoring code against real NHANES patient lab data, fanning in to a synthesis agent for a ranked referral recommendation.
- Backend served live inference with Gemma 4 26B running on-GPU via Ollama on an AMD MI300X droplet (AMD Developer Cloud), with automatic failover to Fireworks GLM 5.2.
- Independently designed and delivered the live product demo and full slide deck presented to hackathon judges.

AI Application & LLM Development *2025–2026*
Python, Gemma 3B, Qwen 3, LangChain, LangGraph, Anthropic API

- Deployed custom Large Language Models locally (including Gemma 3B and Qwen 3) to study end-to-end model setup, inference, and multi-agent system performance.
- Built automated text summarizer AI applications utilizing the Anthropic API, integrating Retrieval Augmented Generation (RAG) and Agent-to-Agent (A2A) architectures for dynamic data processing.
- Ran structured prompt evaluations across model versions and documented findings to track output quality.

Global Hackathons *2025–2026*
International Collaboration, AI Engineering, Cloud Services, Jupyter Notebook

- eGov Hackathon PH: Upcoming participant focused on building civic technology and localized innovations.
- UPLB Code Wars: Competed in a day-long "shadow coding" challenge, solving complex algorithmic problems entirely in Notepad without access to compilers, internet, or AI assistance.

CHIP-8 Emulator *2026*
C/C++, Raylib, System Architecture

- Engineered a robust CHIP-8 emulator from scratch, implementing core CPU instructions and memory management.
- Built a custom visual debugger using Raylib to track system state, registers, and memory for analytical debugging.

Heart Disease Prediction Model *2025*
Python, pandas, Google Colab

- Trained a supervised learning model in Python to predict heart disease from patient features end-to-end.
- Cleaned data and performed feature engineering with pandas, handling missing values and categorical variables.

TECHNICAL SKILLS

Languages: Python, SQL, C, C++, HTML

ML & Data: Feature Engineering, Evaluation Metrics (accuracy, precision, recall, F1, confusion matrix), pandas, Data Preprocessing, Prompt Engineering, Model Deployment

Agentic AI & Orchestration: LangChain, LangGraph, FastAPI, Multi-Agent Systems, Ollama

Methods & Tools: Statistical Analysis, Analytical Debugging, Technical Documentation, Model Evaluation, Google Colab, Git, Linux/CLI, Raylib

Certifications: AMD Multiagent Systems Deployment, CS50 Intro to CS, AI Prompt Engineering, Python, C

Spoken Languages: Filipino (native), English, Latin

LEADERSHIP AND ACTIVITIES

Competitive Debater *2022–2025*
Various National Tournaments, Philippines
Quarterfinalist at Ateneo Peace Debate; competed at UP Diliman Debates, XSDC, and ASDC, honing the ability to present technical arguments clearly under pressure.

Google Developer Group and UP Socomsci *2025–2026*
Coordinated cross-functional teams on operations and event setup; active in student tech communities at UP Manila.

Math Team Captain *2022–2024*
PAREF Southridge School, Muntinlupa, PH
Led tutoring sessions and mentored peers for local math competitions.

AWARDS

DOST Undergraduate Scholar (2025–Present) | Top 5 Finalist, Olymphysics NCR (2025) | 5th Place, Philippine Statistics Quiz NCR (2024)